Curriculum Vitae

Name (ID):	FERNÁNDEZ SAN MIGUEL, ANDRÉS
Organization:	University of A Coruña (UDC)
Date of Birth:	April 20, 1996 Sex: M
Faculty or School:	School of Civil Engineering
Department:	Department of Mathematics
Postal address:	Campus de Elviña s/n, 15071, A Coruña, Spain
Email:	`andres.fernandez.sanmiguel@udc.es`

CURRENT POSITION

Position:	Postdoctoral Researcher
Institution:	University of A Coruña
Starting date:	February, 2026

DEGREES

Degree	School	Date
PhD Civil Engineering	School of Civil Engineering (UDC)	19/02/2026
MSc Civil Engineering	School of Civil Engineering (UDC)	22/03/2022
BSc Civil Engineering	School of Civil Engineering (UDC)	11/04/2019

PhD

Dissertation:	A FEM-based Formulation for Linear and Non-Linear Molecular Elasticity
Center:	School of Civil Engineering. University of A Coruña
Advisors:	Dr. Iván Couceiro Aguiar & Dr. Luis Ramírez Palacios

PREVIOUS SCIENTIFIC-TECHNOLOGICAL ACTIVITIES

Organization	Position	Dates
University of Michigan	PhD Student	01/07/2025 – 31/08/2025
Purdue University	PhD Student	18/08/2023 – 01/10/2023
Univ. of A Coruña	Undergraduate Researcher	16/06/2021 – 30/09/2021
Univ. of A Coruña	Undergraduate Researcher	14/05/2020 – 13/05/2021
POMAR-WATER	Civil Engineer	01/07/2019 – 30/11/2019
IN-CUBED	Civil Engineer Internship	01/07/2018 – 30/09/2018

LANGUAGES (R=Regular, P=Properly, C=Correctly)

Language	Speak	Read	Write
Spanish	C	C	C
English	C	C	C

Date: May 26, 2026 Signature:

All data presented in this Curriculum Vitae are true. Otherwise, the author assumes all responsabilities due to possible mistakes or inaccuracies.

1 PUBLICATIONS

1. **A. Fernández-San Miguel**, I. Couceiro, L. Ramírez, F. Navarrina, **"A first order FEM based formulation for the analysis of molecular structures with bonded interactions."**, *Engineering with Computers*, (2025)

2. **A. Fernández-San Miguel**, I. Couceiro, L. Ramírez, F. Navarrina, **"A Comparative Review of FEM-like Techniques Applied to the Linear Analysis of Molecular Structures."**, *Archives of Computational Mechanics*, (2025)

3. **A. Fernández-San Miguel**, I. Couceiro, L. Ramírez, F. Navarrina, JC. Verduzco, **"Linear and nonlinear elastic analysis of molecular structures using an exact atomistic approach based on geometry and potential energy including non-bonded interactions."**, *Computational Mechanics (Under review)*, (2026)

4. **A. Fernández-San Miguel**, L. Ruano, JJ. Nogueira, L. Ramírez, F. Navarrina, **"Atomistic structural analysis of linear elastic behavior in coronavirus spike proteins."**, *(Under submission)*, (2026)

2 COMMUNICATIONS

1. **A. Fernández San Miguel** et al., **"A FEM-based formulation for Linear and Nonlinear Molecular Elasticity"**, *18th US National Congress on Computational Mechanics (USNCCM18) (Oral presentation)*, Chicago, USA, July 13 - 17, 2025

2. **A. Fernández San Miguel**, F. Navarrina, I. Couceiro, L. Ramírez, **"Matrix analysis of molecular structures: from linear analysis to buckling"**, *16th World Congress on Computational Mechanics (WCCM 2024) (Oral presentation)*, Vancouver, Canada, July 21 - 26, 2024

3. **A. Fernández San Miguel**, F. Navarrina, I. Couceiro, L. Ramírez, **"Matrix analysis of molecular structures: from linear to non linear analysis."**, *9th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS 2024) (Oral presentation)*, Lisboa, Portugal, June 3 - 7, 2024

4. **A. Fernández San Miguel**, F. Navarrina, I. Couceiro, L. Ramírez, **"A New Approach for Vibrational Analysis of Molecular Structures"**, *Congress on Numerical Methods in Engineering (CMN 2022) (Oral presentation)*, Las Palmas de Gran Canaria, Spain, September 12 - 14, 2022

5. L. Ramírez, A. Eirís, L. Edreira, **A. Fernández San-Miguel**, F. J. Fernández-Fidalgo, I. Couceiro, J. París, F. Navarrina, X. Nogueira, I. Colominas, **"A high accurate meshless approach based on SPH method for CFD"**, *Congress on Numerical Methods in Engineering 2022*, Las Palmas de Gran Canaria, Spain, September 12 - 14, 2022

6. I. Colominas, L. Ramirez, X. Nogueira, I. Couceiro, F. Navarrina, J. Paris, F.J. Fernandez-Fidalgo, M.A. Soage, A. Eiris, A. Prieto, L. Edreira, **A. Fernandez San-Miguel**, **"From Mesh to Meshless: A High Accurate SPH method for CFD applications"**, *High-Order Nonlinear Numerical Methods for Evolutionary PDEs: Theory and Applications (HONOM 2022)*, Braga, Portugal, April 4 - 8, 2022

3 PARTICIPATION IN R&D PROJECTS

Project:	**Modelos Numéricos de alta precisión para el desarrollo de una nueva generación de parques Offshore de energía renovable**
Financed by:	Ministerio de Ciencia e Innovación
Participants:	Universidade da Coruña
Total amount:	164.802,00 Euros Period: 2022-2024
Principal Researcher:	Luis Ramírez Palacios & Xesús Antón Nogueira Garea

Project:	**VirionBreak: Cálculo dinámico de la cápside del SARS-CoV-2 para su destrucción por resonancia (COV20/01275)**
Financed by:	Insituto de Salud Carlos III, Ministerio de Ciencia e Innovación
Participants:	Universidade da Coruña
Total amount:	59,250.00 Euros Period: 2020-2021
Principal Researcher:	Fermín Navarrina Martínez

Project:	**Ayudas para la Consolidación y Estructuración de Unidades de Investigación Competitivas del Sistema Universtiario de Galicia, Prof. Colominas Ezponda Nº ED4321X 2018/41, Modalidad de Grupos de Referencia Competitiva**
Financed by:	Consellería de Educación, Universidade e Formación Profesional. Xunta de Galicia
Participants:	Universidade da Coruña
Total amount:	280,000.00 Euros Period: 2019-2021
Principal Researcher:	Ignasi Colominas Ezponda

4 COURSES

1. | | |
   |---|---|
   | Title: | **11th CISM–AIMETA Advanced Course on "Machine Learning for Solid Mechanics"** |
   | Organizer: | International Centre for Mechanical Sciences (CISM) |
   | Place: | Udine, Italy |
   | Date: | September 29 – October 3, 2025 |

2. | | |
   |---|---|
   | Title: | **Density Functional Theory** |
   | Organizer: | École Polytechnique |
   | Platform: | Coursera |
   | Date: | February 19, 2025 |

3. | | |
   |---|---|
   | Title: | **NL24 – Nonlinear Computational Solid & Structural Mechanics** |
   | Organizer: | University of Pavia |
   | Place: | Pavia, Italy |
   | Date: | May 06–10, 2024 |

5 OTHER RELEVANT MERITS

- J-1 Visiting Scholar from 01/07/2025 to 31/08/2025 at University of Michigan (Computational Vascular Biomechanics Lab)

- J-1 Visiting Scholar from 18/08/2023 to 01/10/2023 at Purdue University (Strachan Lab)
- Member of SEMNI (Spanish Society of Computational Mechanis and Computational Engineering).
- ACADAR-MCVALNERA Award for the best project in port and coastal engineering.
- Puerto Deportivo Sada Award for the best study and innovative project in port engineering.